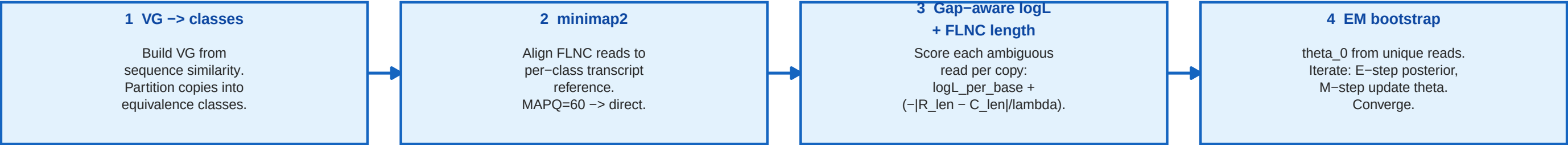


Pipeline + recovery: multi-copy FLNC quantification works end-to-end

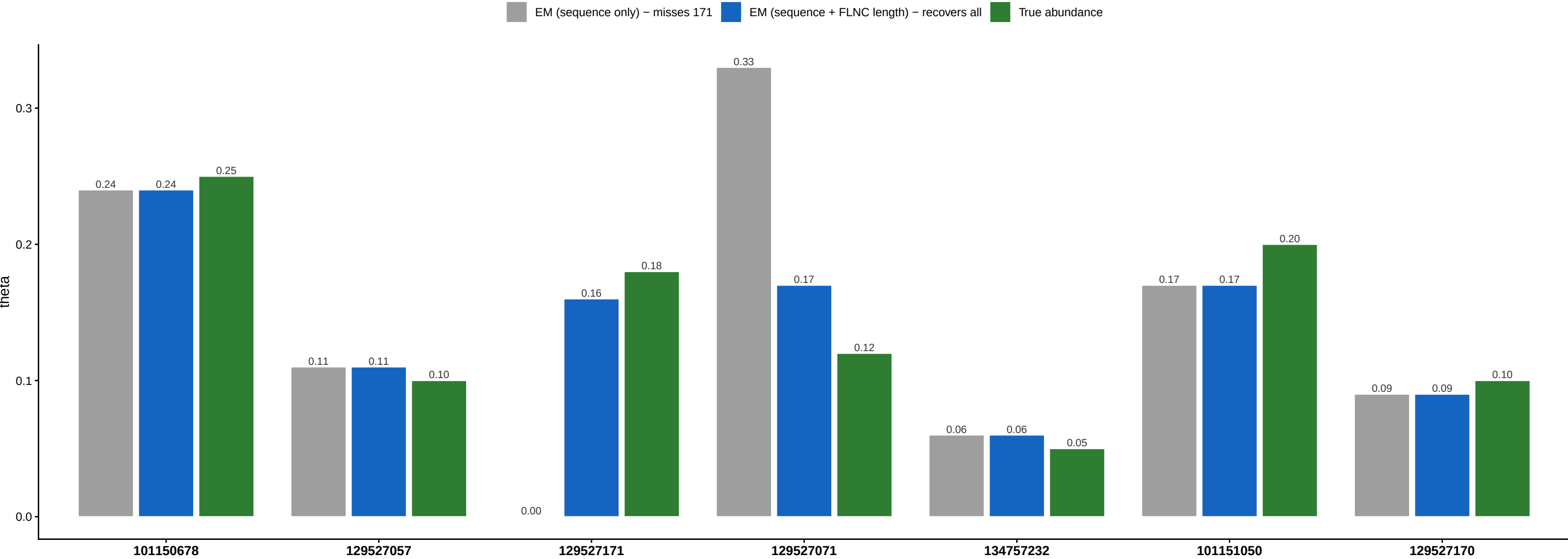
GOLGA8I as the worked example. Sequence-only EM fails on 171 vs 071 (paralog repeat sequence). FLNC length consistency closes that gap.

A Pipeline above the resolution floor



B Result: GOLGA8I library, 100 simulated FLNC reads

Sequence-only EM: 83 unique reads, 171 absorbed into 071. Sequence + FLNC length: 100 unique reads, all copies recovered.



BOTTOM LINE Multi-copy gene family read-to-copy assignment IS resolvable

Down to the VG equivalence-class floor (Theorem 1, set by identity threshold).

Above the floor: minimap2 + class-aware MAPQ + gap-aware HMM + FLNC length + EM bootstrap recovers true abundances (here: 100% of 100 simulated reads).